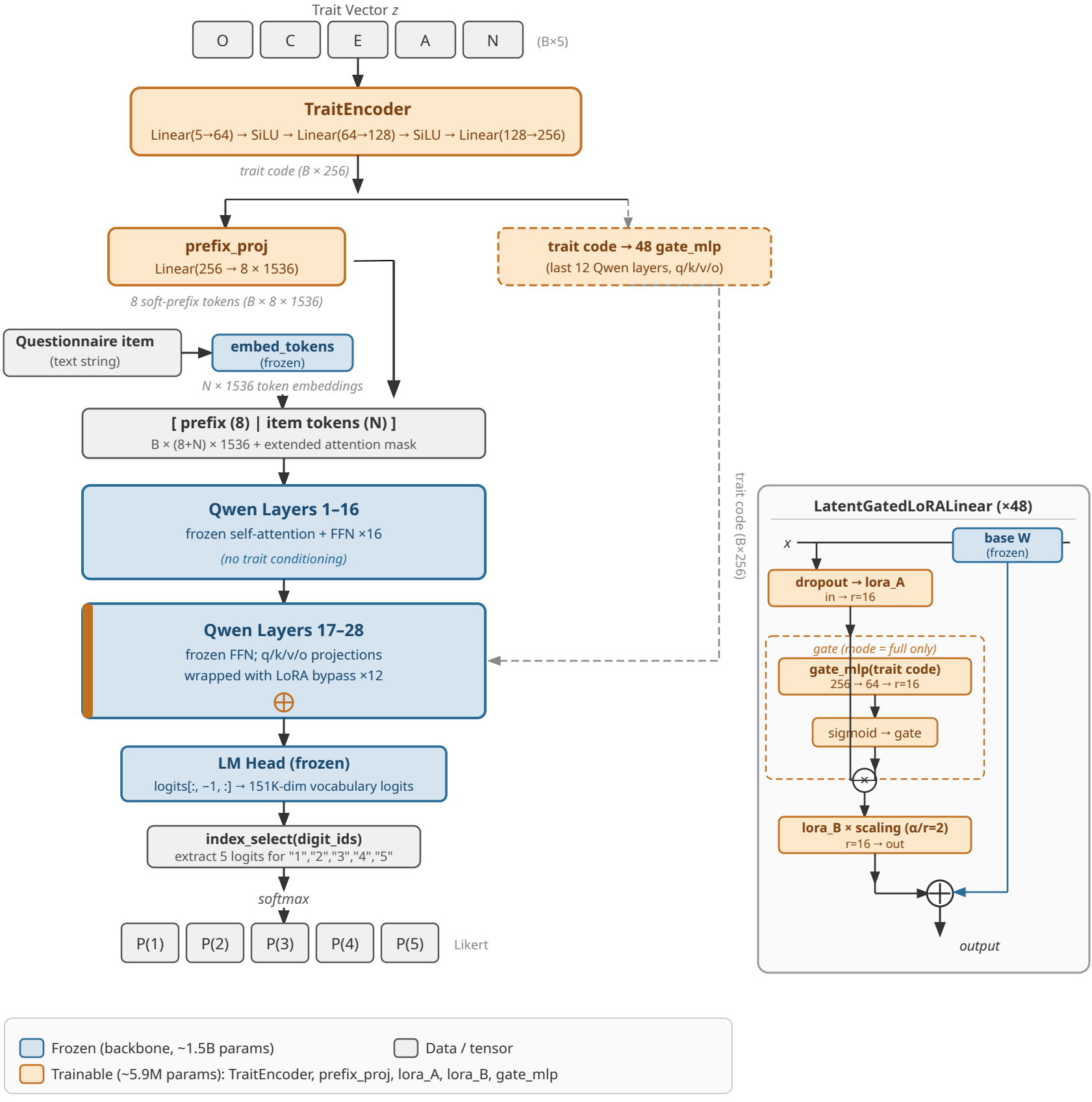

Backbone: Qwen2.5-1.5B-Instruct (frozen). LoRA: r=16, α=32 on q/k/v/o of final 12 layers (48 modules).
Soft prefix: 8 learned tokens prepended to input embeddings. Answer readout: softmax over digit tokens "1"–"5" at final position.